Physical Chemistry (I)

Lecture 09

1차 중간고사 안내 및 복습

1차 중간고사 안내

- 시간: 4월 3일(목) 오전 10:30~11:30
- 장소: 화학관 315호(홀수 학번), 112호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

- 1장 “열역학의 기본 개념들”
- 2장 “기체의 성질”
- 3장 “열역학 제1법칙”
- 4장 “엔트로피”

시험 문제

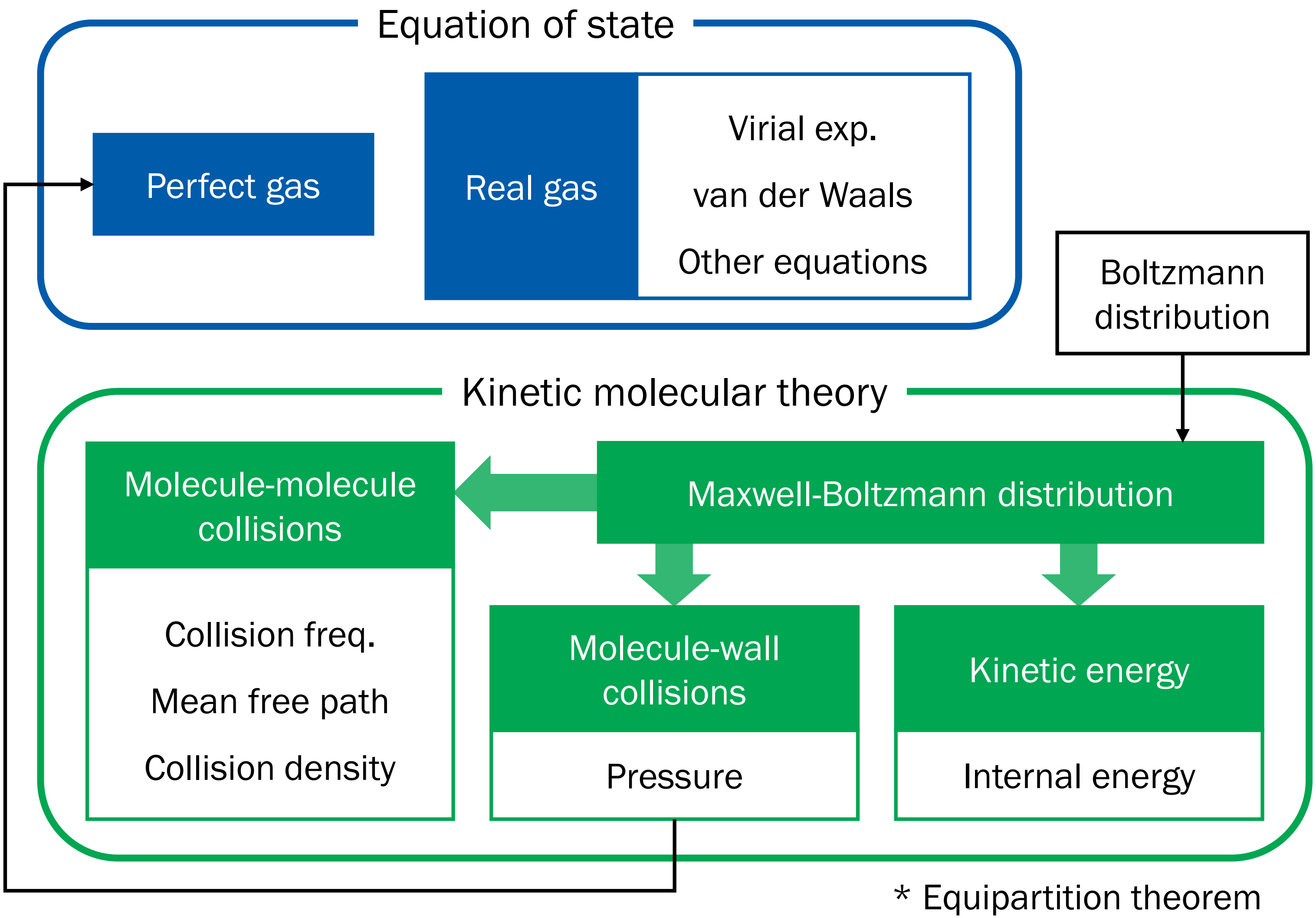
총 여섯 문제

- 과제 문제 혹은 그 변형(3문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(3문제)
  - 강의 자료 기준

Concepts in Chpt 1

- System and surroundings
- Classification of systems
- System properties: extensive vs. intensive
- Equilibria: mechanical, diffusive, and thermal
- The zeroth law of thermodynamics
- State of a system
- State change

Chapter 2



Equation of state
Perfect gas
Real gas
Virial exp.
van der Waals
Other equations
Boltzmann distribution
Kinetic molecular theory
Molecule-molecule collisions
Collision freq.
Mean free path
Collision density
Maxwell-Boltzmann distribution
Molecule-wall collisions
Pressure
Kinetic energy
Internal energy
* Equipartition theorem

Concepts and Derivations in Chpt 2

- [C] Perfect gas vs. real gas
- [C] Equation of state, perfect gas law
- [C] Kinetic molecular theory
- [C] Boltzmann distribution
- [D] Maxwell-Boltzmann distribution
- [D] Characteristic speeds
- [D] Collision frequency, mean free path, collision density
- [D] Pressure

Concepts and Derivations in Chpt 2

- [D] Internal energy
- [C] Equipartition theorem
- [C] Compression factor
- [C] Virial expansion
- [C] Boyle temperature
- [C, D] van der Waals equation
- [D] van der Waals and virial
- [C] Other real gas equations: no need to memorize

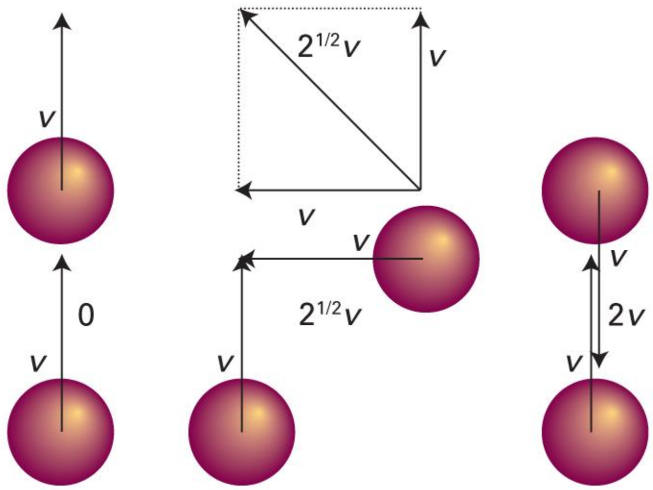
적분표의 활용

$$\boxed{\int_0^\infty x e^{-kx^2}\,\mathrm{d}x = \frac{1}{2k}}$$

$$\int_0^\infty v e^{-Mv^2/2RT}\,\mathrm{d}v = ?$$

$$\int_{-\infty}^\infty v e^{-Mv^2/2RT}\,\mathrm{d}v = ?$$

$\sqrt{2}$는 어디에서?



(Image: Atkins *et al*., ***Atkins' Physical Chemistry***, 12th ed., Figure 1B.7)

Chapter 3 (1)

First Law of Thermodynamics

$$\mathrm{d}U = \delta q + \delta w$$

Work	$\delta w = -p_{\mathrm{ex}}\mathrm{d}V + \delta w_{\mathrm{add}}$ • Constant-p change • Reversible change
Heat	$\delta q = C\mathrm{d}T$ • Constant-V change $C_V = (\partial U/\partial T)_V$

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Differential form of U

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T$$

- Internal pressure π_T
- Response functions α, κ_T

Chapter 3 (2)

Thermodynamics	Mathematics
State function	Exact differential
Path function	Inexact differential

Enthalpy

$$H = U + pV$$

$$\mathrm{d}H = \delta q_p$$

Differential form of U

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T$$

Differential form of H

$$\mathrm{d}H = \left(\frac{\partial H}{\partial p}\right)_T \mathrm{d}p + \left(\frac{\partial H}{\partial T}\right)_p \mathrm{d}T$$

$$C_V = (\partial U/\partial T)_V$$

C_V and C_p

- Perfect gas
- Adiabatic change: $\delta q = 0$

$$C_p = (\partial H/\partial T)_p$$

Chapter 3 (3)

Enthalpy

$H = U + pV$

Standard enthalpy $H^{\ominus}$

Standard enthalpy change $\Delta H^{\ominus}$

Heat capacity

$C_p = (\partial H / \partial T)_p$

Hess's law

Kirchhoff's law

Standard reaction enthalpy $\Delta_r H^{\ominus}$

$$\Delta_r H^{\ominus} = \sum_{\text{product } i} \nu(i) H_m^{\ominus}(i) - \sum_{\text{reactant } j} \nu(j) H_m^{\ominus}(j)$$

Standard enthalpy of formation $\Delta_f H^{\ominus}$

$$\Delta_r H^{\ominus} = \sum_{\text{product } i} \nu(i) \Delta_f H^{\ominus}(i) - \sum_{\text{reactant } j} \nu(j) \Delta_f H^{\ominus}(j)$$

Concepts and Derivations in Chpt 3

- [C] First law of thermodynamics
- [D] Expansion work
  - Isochoric and isobaric change
  - Free expansion
  - Reversible (perfect gas, isothermal) change
- [D] Heat
  - Isochoric and isobaric change
  - Internal energy and isochoric heat capacity

Concepts and Derivations in Chpt 3

- [C] State and path functions
- [C, D] Differential form of internal energy
- [C] Response functions
- [D] Derivation of $(\partial U/\partial T)_p$
- [C, D] Enthalpy
  - Reversible isobaric change
  - Enthalpy and isobaric heat capacity
- [C, D] Differential form of enthalpy

Concepts and Derivations in Chpt 3

- [D] Heat capacities for perfect gas
- [C, D] Adiabatic change
  - Work for perfect gas
  - Reversible, perfect-gas relations: V-T and V-p
- [C] Standard enthalpy
  - Standard reaction enthalpy
  - Standard enthalpy of formation
- [C] Hess's law
- [D] Kirchhoff's law

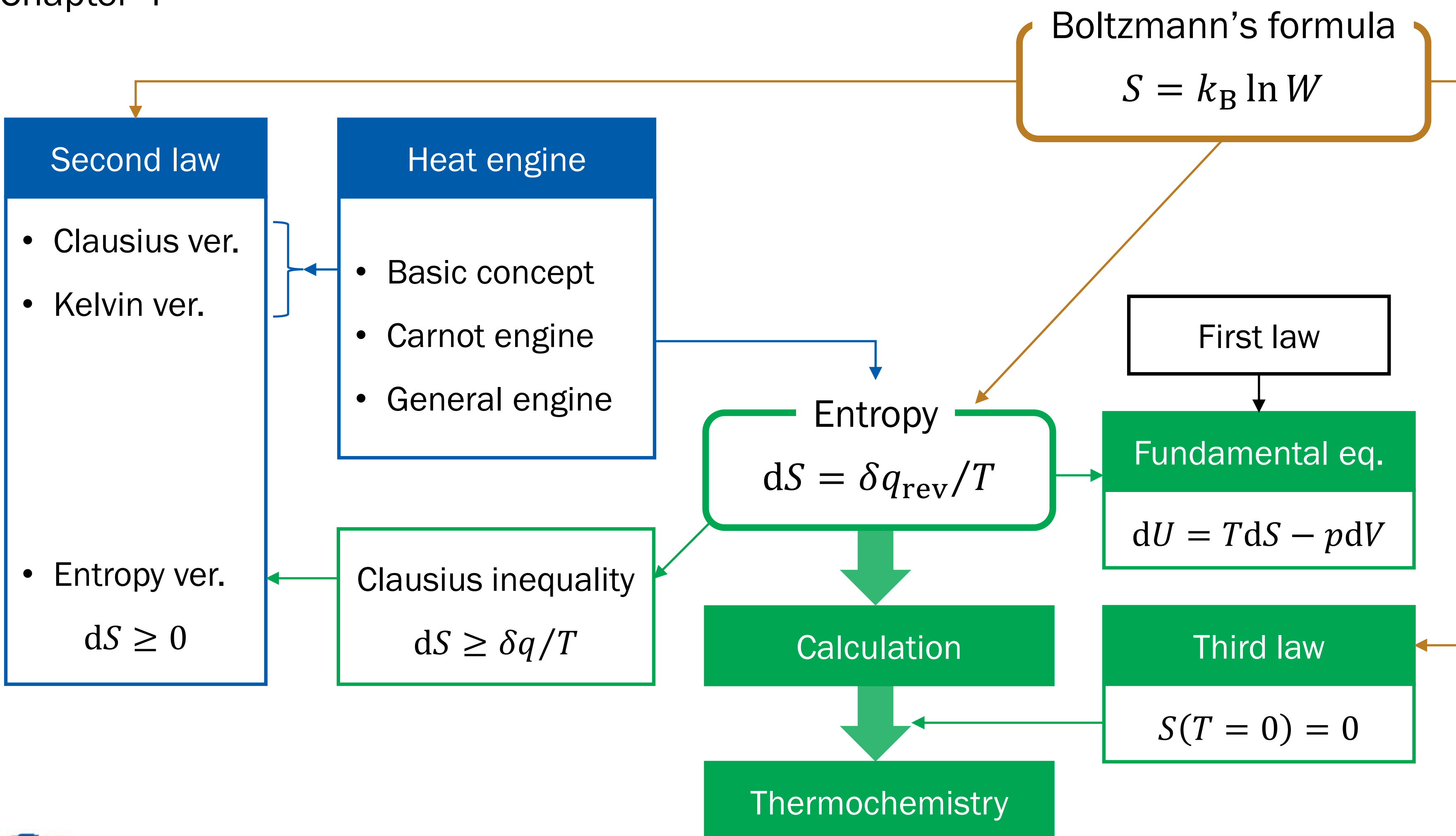
가역 변화

	$w = -\int p\,\mathrm{d}V$	q	ΔU
등압($\mathrm{d}p = 0$)	$-p\Delta V$		
등적($\mathrm{d}V = 0$)	0		
등온($\mathrm{d}T = 0$)			0
단열($\delta q = 0$)		0	

완전 기체의 가역 변화

“열 용량 일정”	$w = -\int p \, \mathrm{d}V$	q	$\Delta U = C_V \Delta T$
등압($\mathrm{d}p = 0$)	$-p\Delta V$	$C_p\Delta T$ $= C_V\Delta T + p\Delta V$	$C_V\Delta T$
등적($\mathrm{d}V = 0$)	0	$C_V\Delta T$	$C_V\Delta T$
등온($\mathrm{d}T = 0$)	$-nRT\ln(V_f/V_i)$	$nRT\ln(V_f/V_i)$	0
단열($\delta q = 0$)	$C_V\Delta T$	0	$C_V\Delta T$

Chapter 4



Boltzmann's formula
$S = k_B \ln W$
Second law
Clausius ver.
Kelvin ver.
Entropy ver.
$dS \geq 0$
Heat engine
Basic concept
Carnot engine
General engine
First law
Entropy
$dS = \delta q_{rev}/T$
Fundamental eq.
$dU = TdS - pdV$
Clausius inequality
$dS \geq \delta q/T$
Calculation
Third law
$S(T = 0) = 0$
Thermochemistry

Concepts and Derivations in Chpt 4

- [C] Heat engine
- [C, D] The second law of thermodynamics
- [C] Engine efficiency
- [C, D] Carnot cycle and Carnot efficiency
- [C] Generalization of Carnot efficiency
- [C] Entropy
- [C, D] Clausius inequality
  - Entropy version of the second law
  - Entropy and spontaneity

Concepts and Derivations in Chpt 4

- [C, D] Fundamental equation of thermodynamics
- [D] Entropy calculations
  - Heat transfer
  - Entropy change of surroundings
  - Expansion
  - Phase transition
  - Heating
- [C] Nernst heat theorem and the third law

Concepts and Derivations in Chpt 4

- [C] Thermochemistry of entropy
  - Standard entropy
  - Standard reaction entropy
- [C] Entropy as an index of exhaustion
- [C] Boltzmann’s formula
- [D] Applications of Boltzmann’s formula
- [C] Reinterpretations of the laws
- [C, D] Independent spin model